

RELATORIO DE ANALISE - MODELO VARMAX

Hiperparametros do Modelo:

```
p      (endog_lag)           : 1
x_lag (exog_lag)           : 1
r      (exog_real_degree)   : 1
i      (exog_img_degree)    : 2
```

Equacao do Modelo:

$$Y(t) = AR(1..1) + X_{real}^{(1..1)} + X_{img}^{(1..2)} \text{ com lag}(1..1)$$

Base de Dados:

```
Tamanho      : 483830
N° Replicações : 10
Randomizada   : True
Seed          : 42
```

Resumo Estatístico:

Statespace Model Results

Dep. Variable:	['Yreal', 'Yimg']	No. Observations:	290298
Model:	VARX(1)	Log Likelihood	-591295.944
	+ intercept	AIC	1182633.888
Date:	Sat, 09 May 2026	BIC	1182856.040
Time:	15:05:00	HQIC	1182698.232
Sample:	0		
	- 290298		
Covariance Type:	opg		

METRICAS DO MODELO

Hiperparametros: p=1, x_lag=1, r=1, i=2

Metricas Gerais:

RMSE Medio	: 0.532619
AIC	: 919964.7982
BIC	: 920483.1527

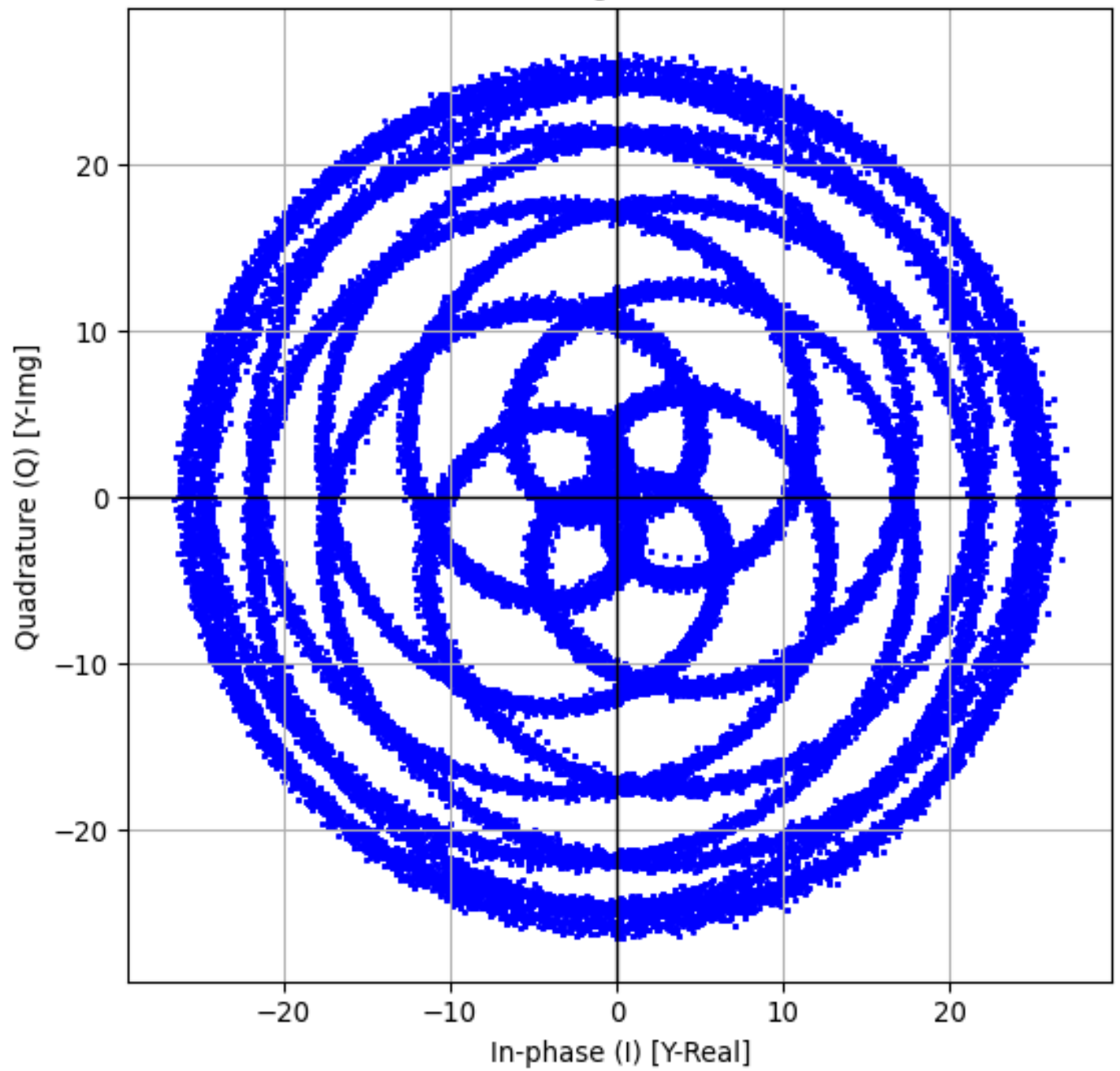
Componente Real (Yreal):

MSE	: 0.283116
RMSE	: 0.532087
MAE	: 0.439994
R2	: 0.998332
Adj R2	: 0.998332
STD	: 0.532087

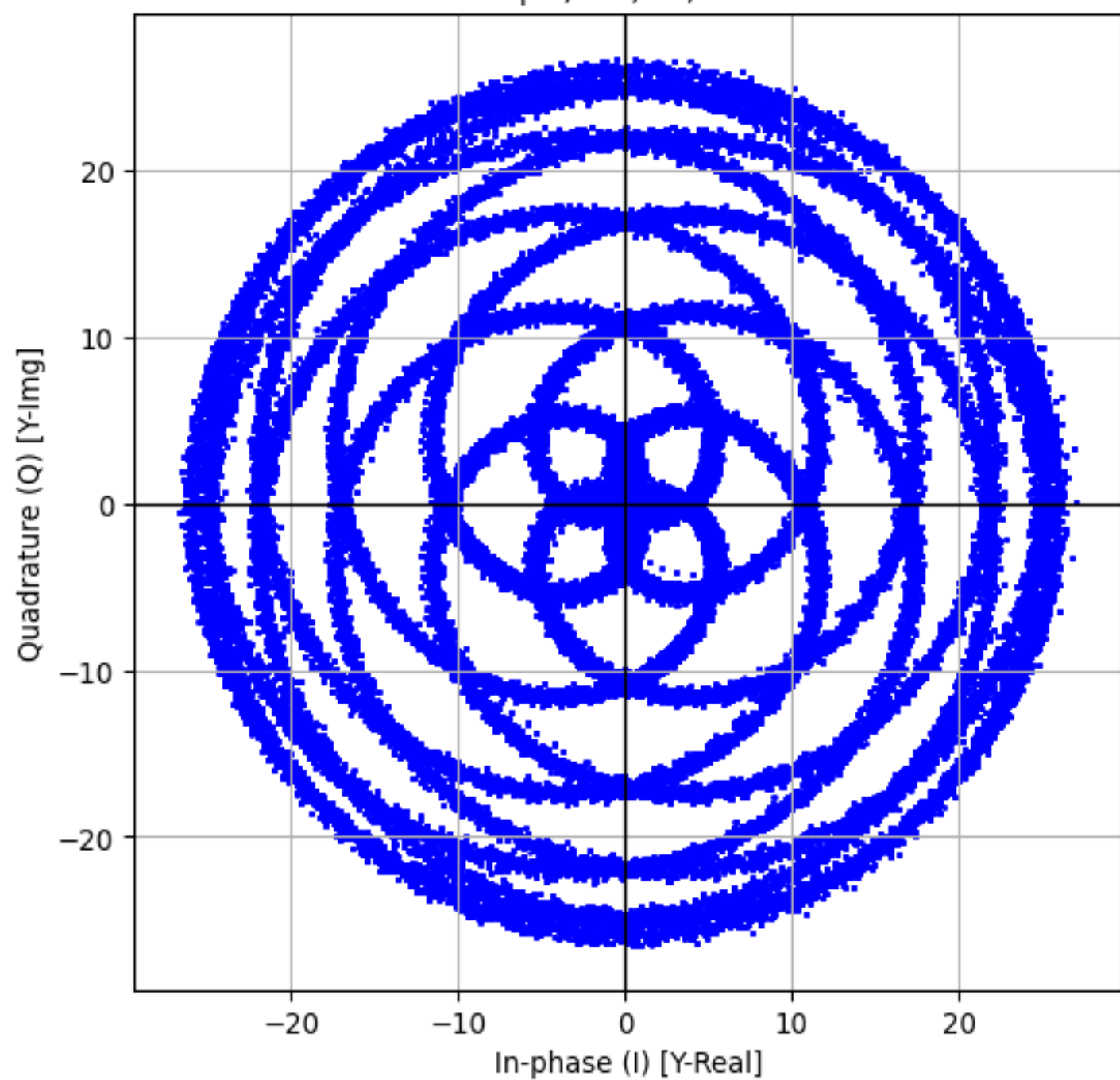
Componente Imaginaria (Yimg):

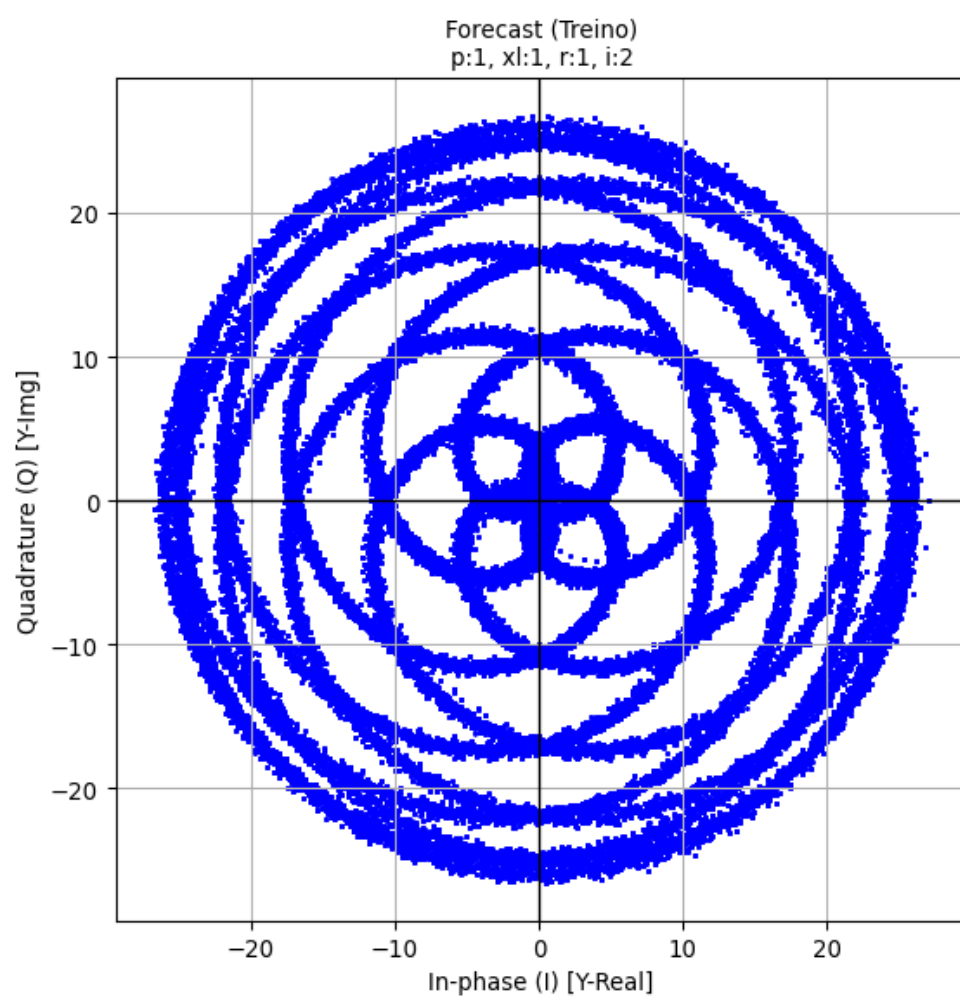
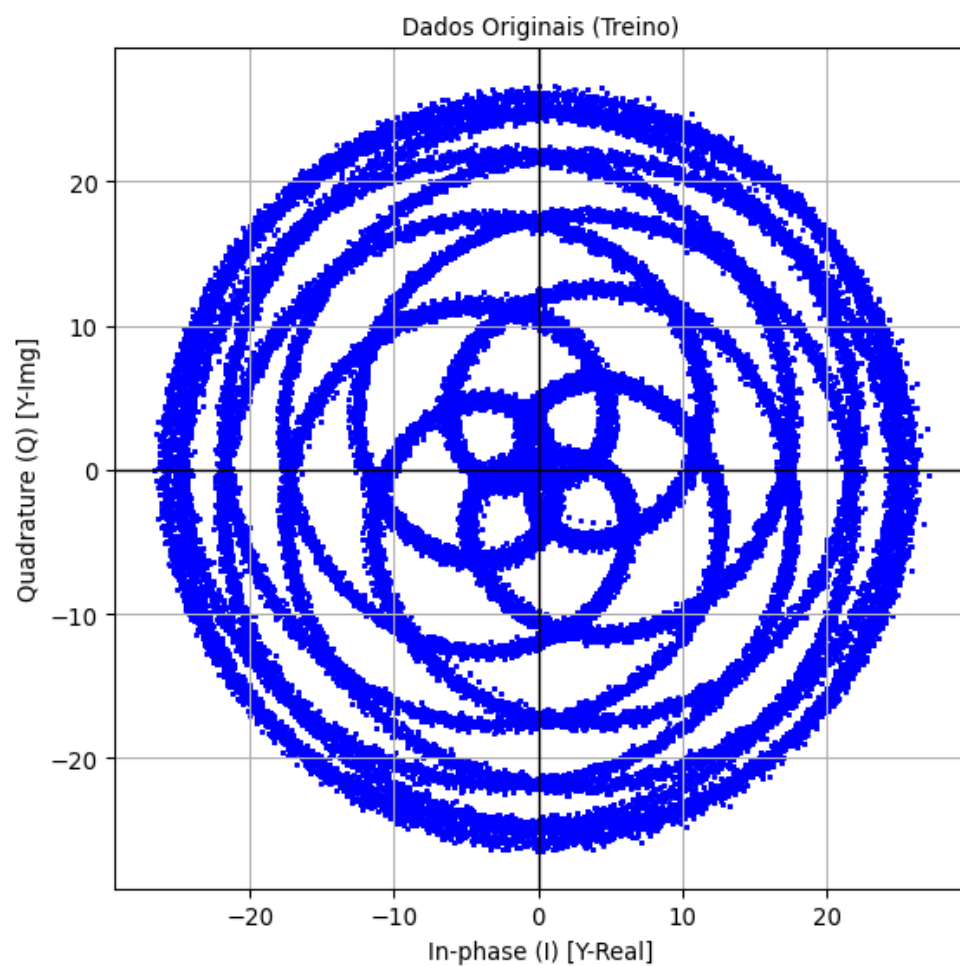
MSE	: 0.284251
RMSE	: 0.533152
MAE	: 0.441164
R2	: 0.998321
Adj R2	: 0.998321
STD	: 0.533149

Dados Originais (Treino)

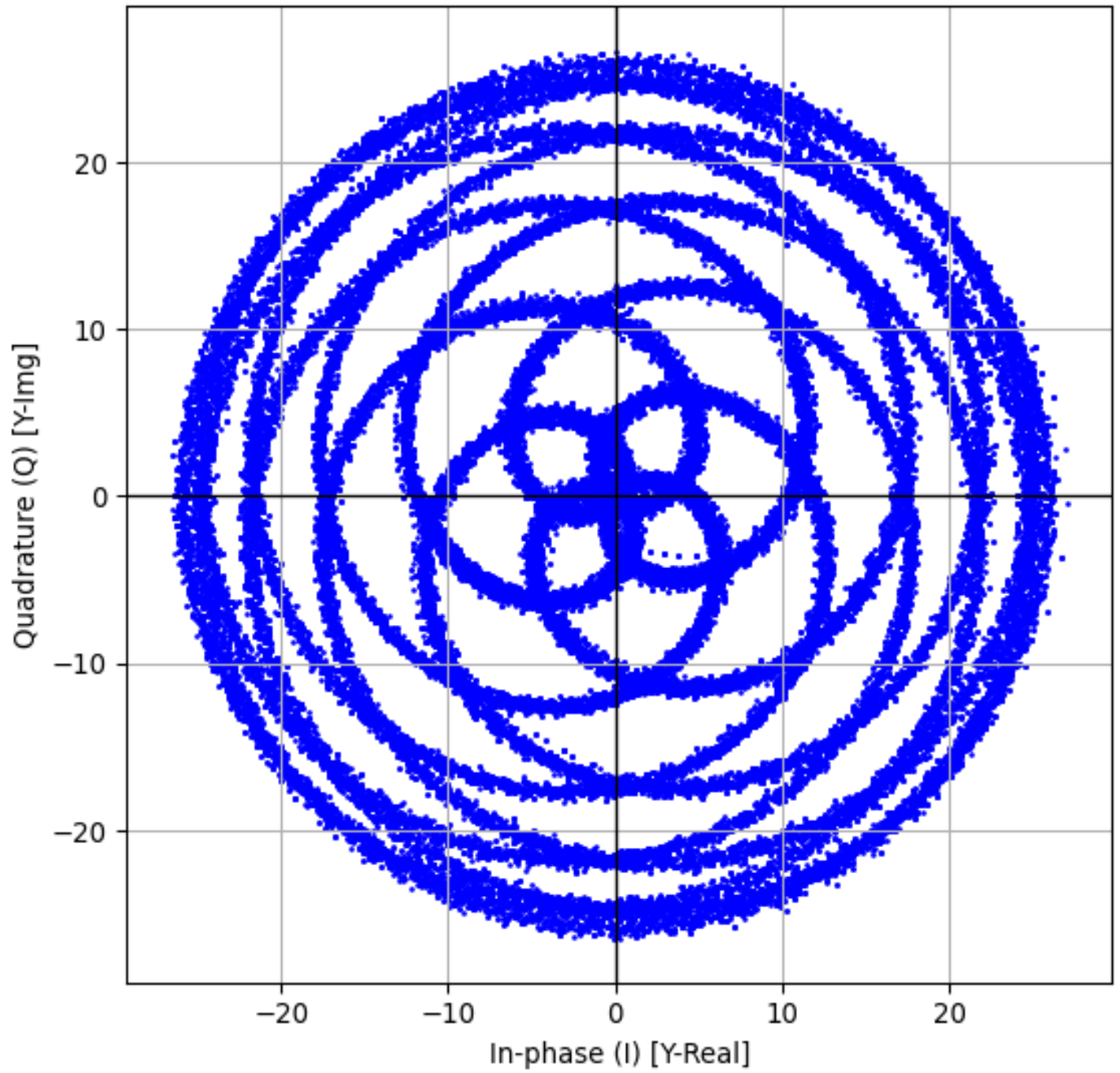


Forecast (Treino)
p:1, xl:1, r:1, i:2

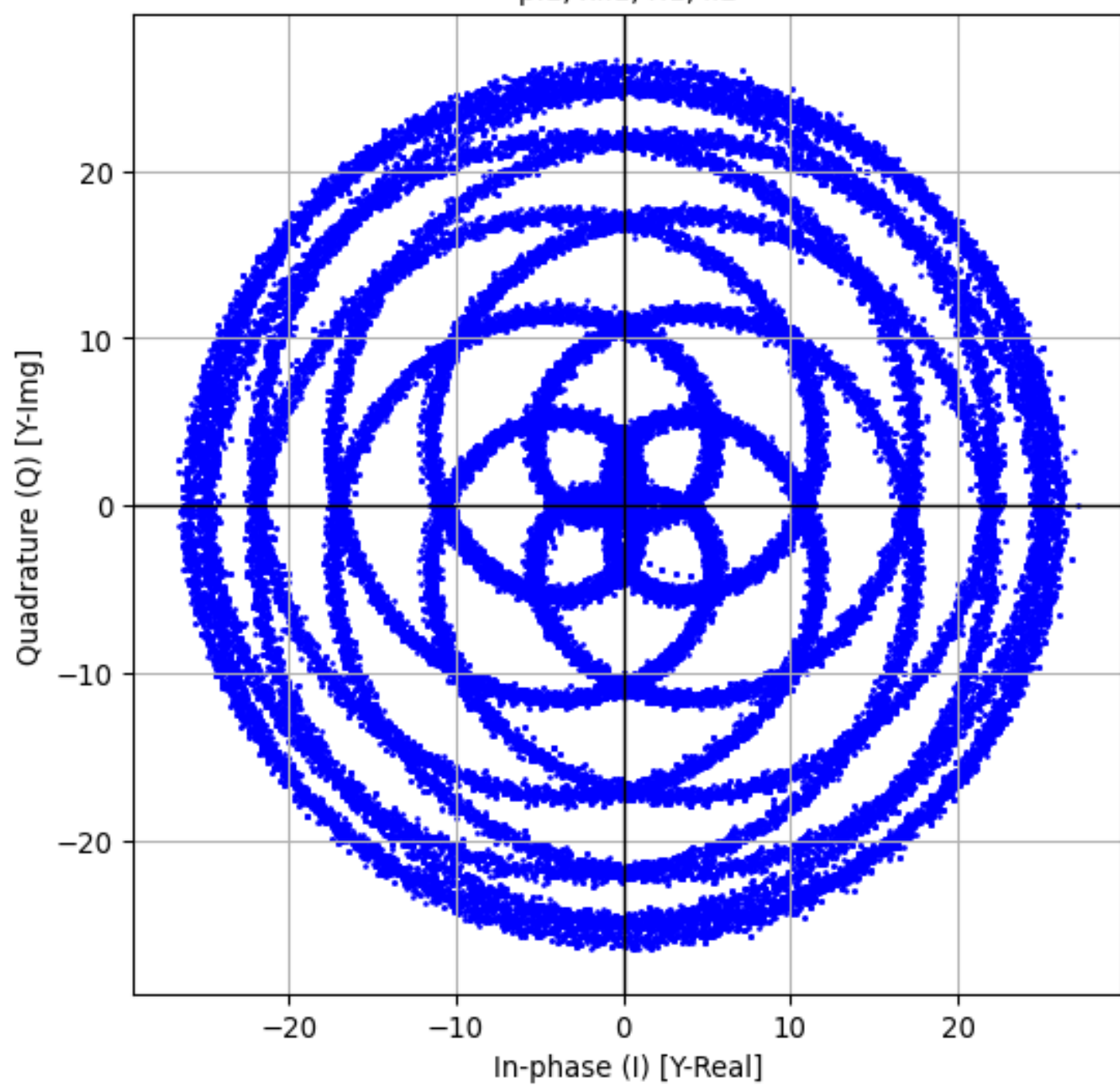


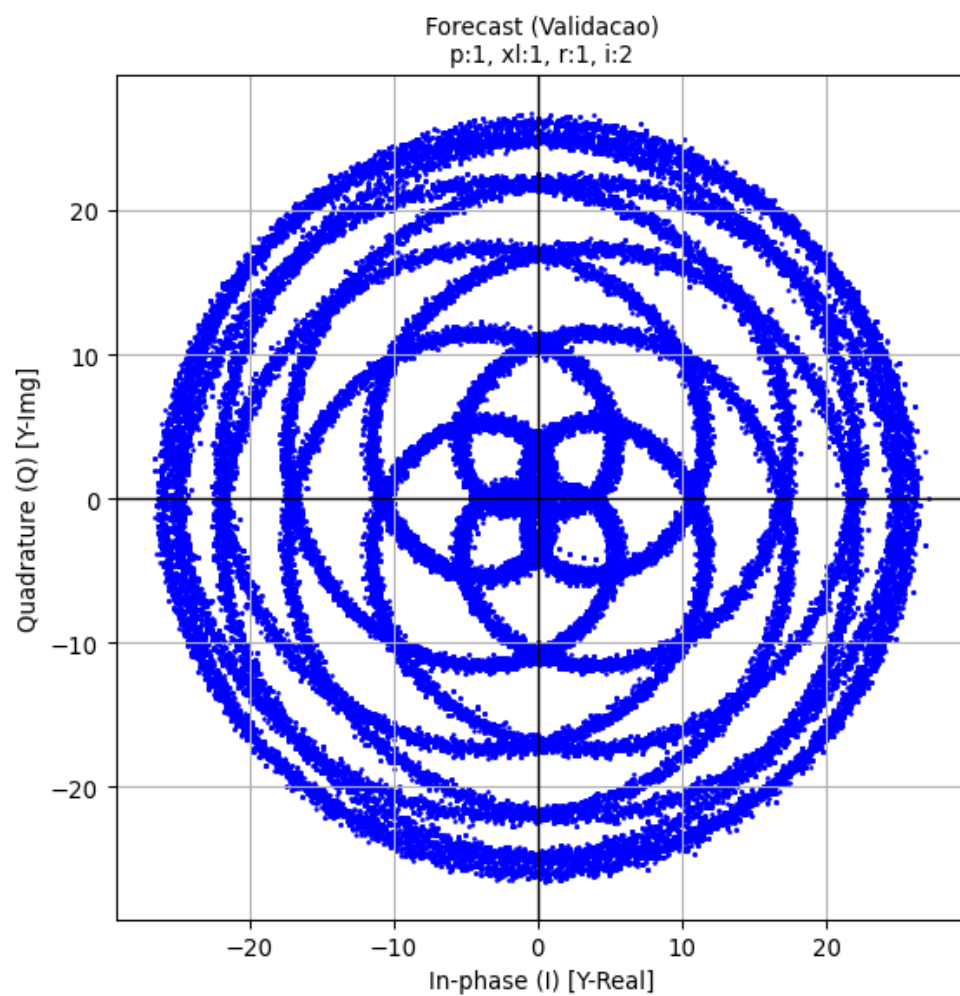
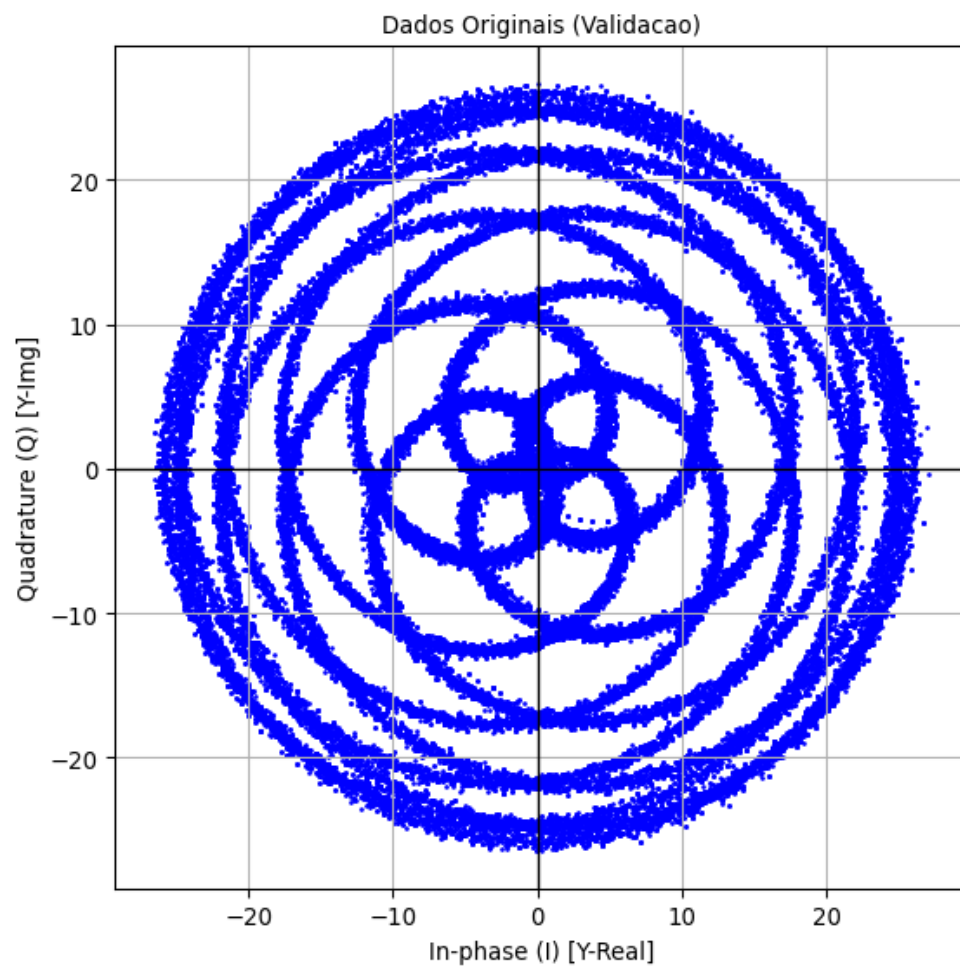


Dados Originais (Validacao)

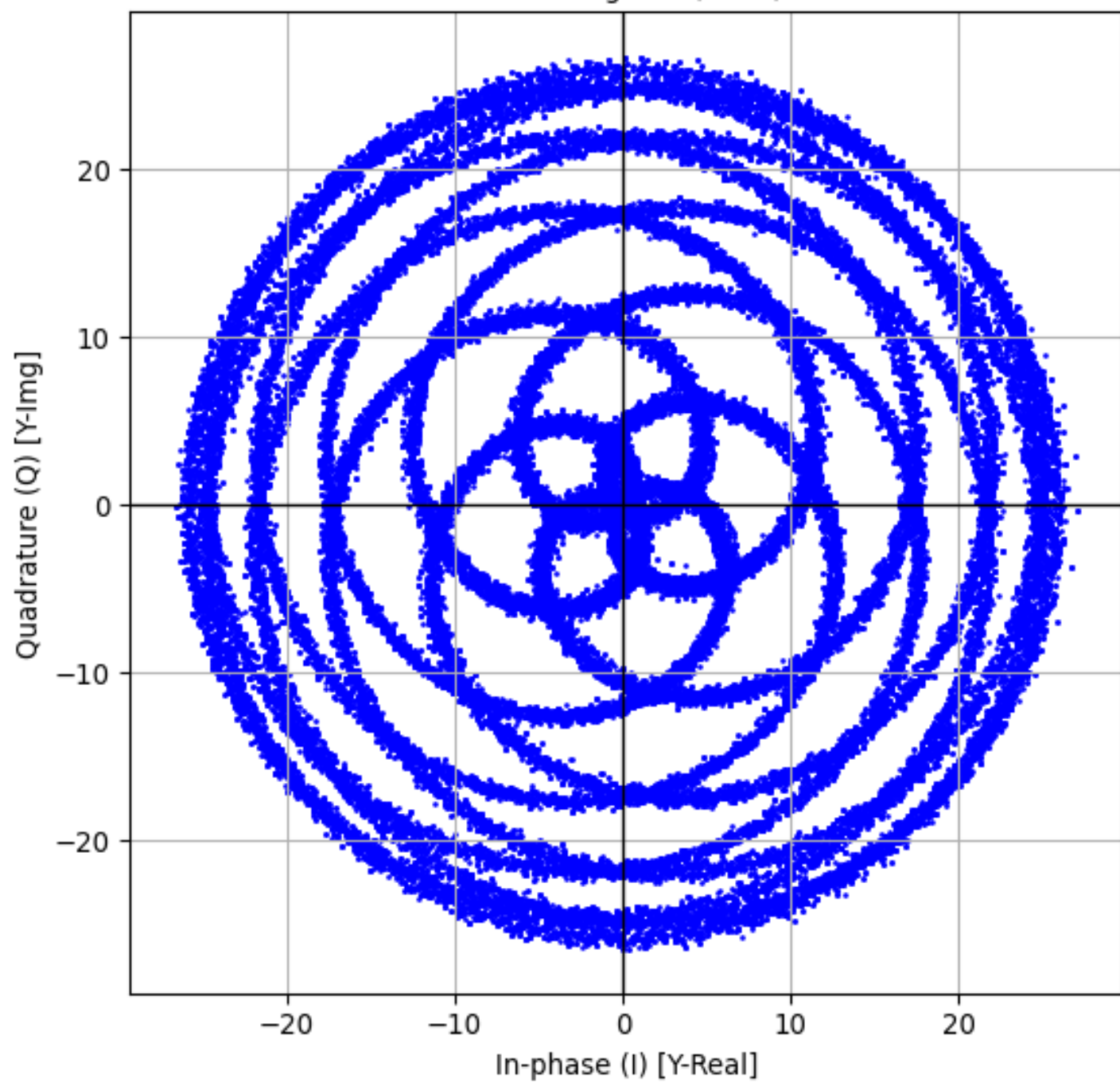


Forecast (Validacao)
p:1, xl:1, r:1, i:2

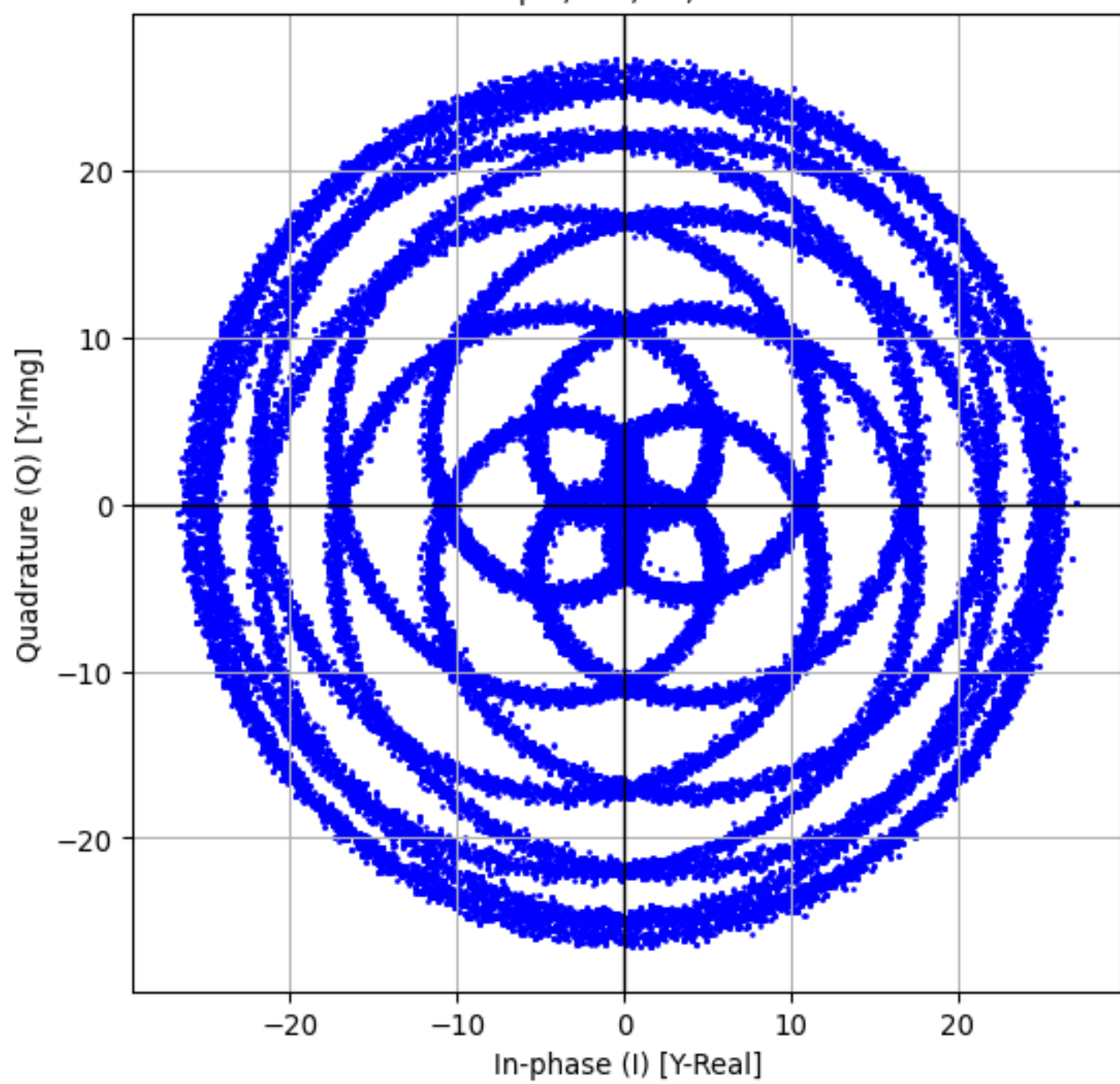




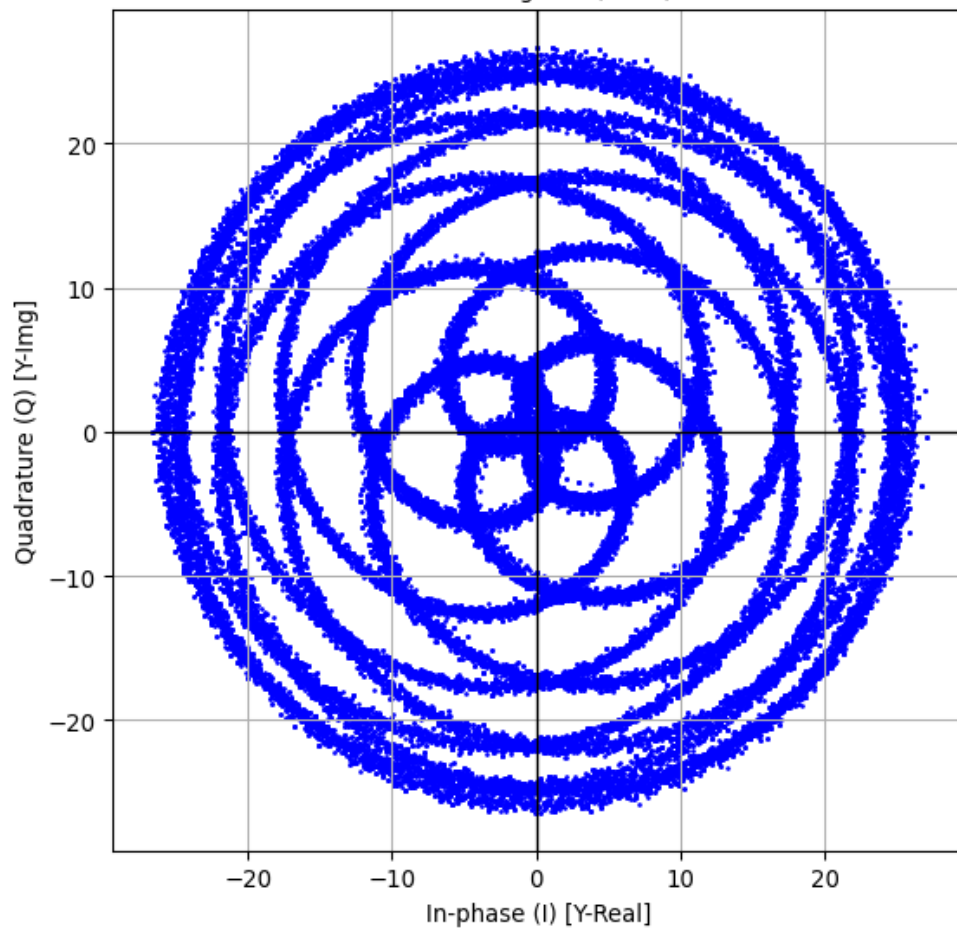
Dados Originais (Teste)



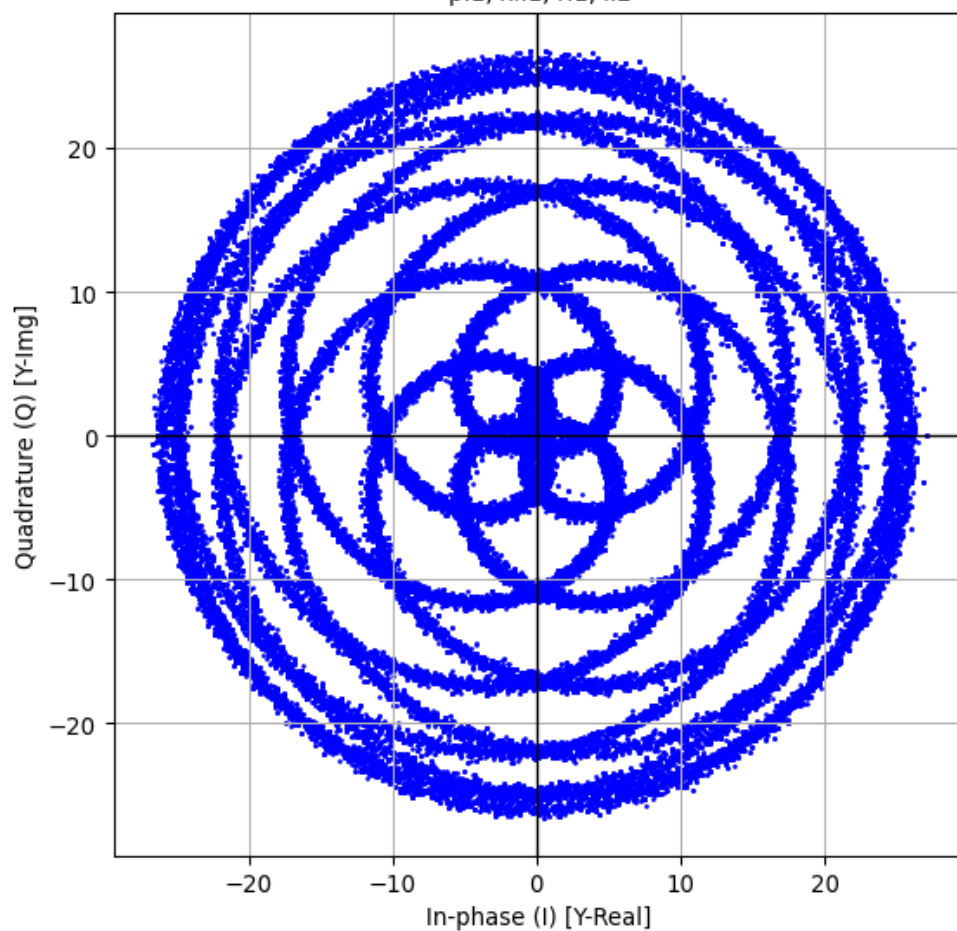
Forecast (Teste)
p:1, xl:1, r:1, i:2



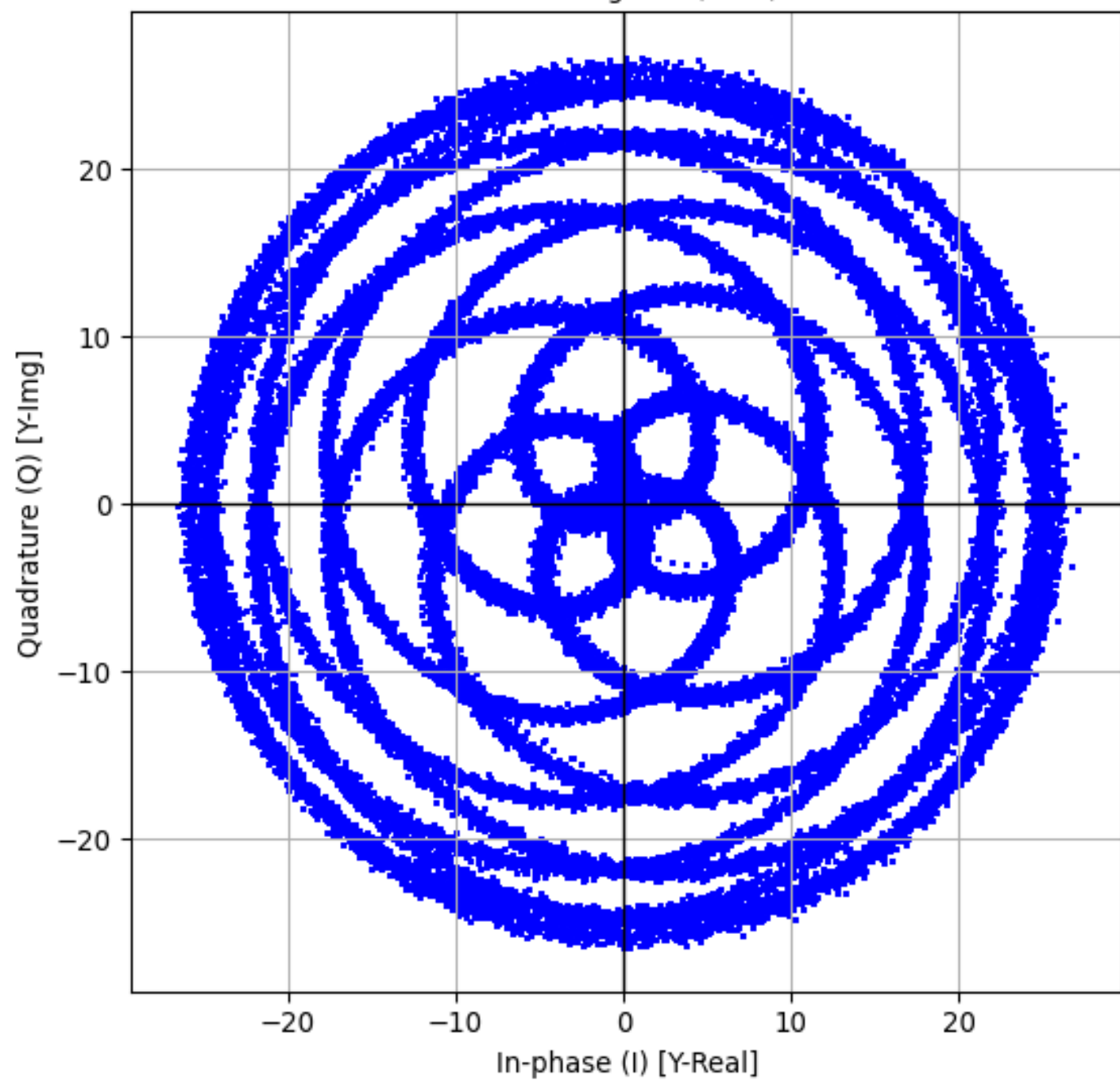
Dados Originais (Teste)



Forecast (Teste)
p:1, xl:1, r:1, i:2



Dados Originais (Total)



Forecast (Total)
p:1, xl:1, r:1, i:2

